

The µOS++ test catalogue

Every test the harness registers, per architecture, platform and board: what each test does, whether it runs the SMP scheduler or a single core, how it is run (QEMU, host or the real board) and, for the board runs, which debug probe.

The sources of truth are the code, not other documents:

What	Where
which cases exist	tests/platforms/<platform>/CMakeLists.txt; for port platforms, the directories under <port>/test/<board>/ (globbed at configure time)
per-test knobs (core count, HWD_ONLY, sources)	<port>/test/<board>/tests.cmake
board facts (CPU, core count, QEMU linker)	<port>/test/boards/<board>/board.cmake
probe	<port>/test/boards/<board>/openocd*.cfg and hw.sh; for the upstream platforms, the openocd command in tests/platforms/<platform>/CMakeLists.txt
xpm actions	tests/package.json

The ports are the sibling repositories micro-os-plus-iii-aarch32.git, -aarch64.git, -cortexm.git and -posix-arch.git; the harness is micro-os-plus-iii-smp.git/tests/.

1. Legend

Variant (the suffix of the CTest name):

Suffix	Runs on	Runner
-qemu	QEMU	test_smpl/run-qemu.sh, or QEMU directly (Cortex-M)
-host	the Linux host (POSIX port)	

Suffix	Runs on	Runner
-hwd	the real board, through OpenOCD	test_smpl/run-host.sh, or the executable directly the board's hw.sh (then test_smpl/run-hw.sh on the Pis)
-test (no variant)	harness-only platforms: QEMU, or the board through OpenOCD	the command in the platform's CMakeLists.txt

-hwd cases carry the hwd label and are excluded from `xpm run test` (`ctest -LE hwd`). They are run one at a time, one per power cycle, with OpenOCD only (no GDB); the verdict is the `RESULT: PASS|FAIL` line, read from semihosting and mirrored on the UART.

Scheduling (the “Mode” column below):

Mode	Meaning
SMP n	kernel built with <code>OS_USE_SMP_SCHEDULER</code> , <code>OS_NCPU = n</code> , all n cores scheduling
SMP n, 1 active	SMP build, but the test keeps the secondaries parked
SMP ×1	SMP scheduler code paths at <code>OS_NCPU = 1</code> (single-CPU board)
single	the kernel's non-SMP branch (<code>OS_USE_SMP_SCHEDULER</code> undefined), one core
bare	no kernel at all: a bare-metal probe

2. Platforms at a glance

Platform (xpm config prefix)	Port	Board / machine	CPU	Cores	Variants	Probe
aarch32-rpi-zero-2w	aarch32	Raspberry Pi Zero 2 W (BCM2837)	4× Cortex- A53, AArch32	4	qemu, hwd	J-Link (default) or Olim
aarch32-rpi3b	aarch32	Raspberry Pi 3 B (BCM2837)	4× Cortex- A53, AArch32	4	qemu, hwd	J-Link (default) or Olim

Platform (xpm config prefix)	Port	Board / machine	CPU	Cores	Variants	Probe
aarch32-luckfox-lyra	aarch32	Luckfox Lyra B (RK3506)	3× Cortex-A7 (+ a Cortex-M0 not scheduled)	3	hwd only	WCH- Link, CMSIS DAP
aarch64-rpi-zero-2w	aarch64	Raspberry Pi Zero 2 W (BCM2837)	4× Cortex-A53, ARMv8-A	4	qemu, hwd	J-Link (default) or Olimex
aarch64-rpi3b	aarch64	Raspberry Pi 3 B (BCM2837)	4× Cortex-A53, ARMv8-A	4	qemu, hwd	J-Link (default) or Olimex
cortexm-pico2	cortexm	Raspberry Pi Pico 2 (RP2350)	2× Cortex-M33	2	qemu (6 tests), hwd	any CMSIS DAP
cortexm-pico2-pizero	cortexm	Pi-Zero RP2350B, 16 MB flash	2× Cortex-M33	2	hwd only	XV-Link CMSIS DAP
cortexm-pico2-rp2350b-psram	cortexm	WeAct RP2350B, 16 MB flash + 8 MB PSRAM	2× Cortex-M33	2	qemu (2 tests), hwd	CMSIS DAP c251:f0
cortexm-nucleof411	cortexm	ST Nucleo-F411RE	1× Cortex-M4F	1	hwd only	on-board ST-Link v2.1
cortexm-weactf411	cortexm	WeAct Studio F411CE	1× Cortex-M4F	1	hwd only	DAPLink CMSIS DAP (or WCH- Link)
cortexm-weactf412	cortexm	WeAct Studio F412RE	1× Cortex-M4F	1	hwd only	ST-Link
native	posix-arch	the Linux host	host threads as CPUs	4 (-DNCPU)	host	—
2xcortex-m33	cortexm	QEMU mps2-an521	2× Cortex-M33	2	QEMU	—
pico2-1cpu	cortexm	QEMU mps2-an505	1× Cortex-M33	1	QEMU	—

Platform (xpm config prefix)	Port	Board / machine	CPU	Cores	Variants	Probe
qemu-cortex-m0	cortexm	QEMU mps2-an385	Cortex-M3 (the platform passes -- cpu cortex-m3, not an M0)	1	QEMU	—
qemu-cortex-m3	cortexm	QEMU mps2-an385	Cortex-M3	1	QEMU	—
qemu-cortex-m4f	cortexm	QEMU mps2-an386	Cortex-M4	1	QEMU	—
qemu-cortex-m7f	cortexm	QEMU mps2-an500	Cortex-M7	1	QEMU	—
nucleo-f411re	harness (upstream)	ST Nucleo- F411RE	Cortex- M4F	1	board	ST-Link (interface stlink- dap.cfg)
nucleo-f767zi	harness (upstream)	ST Nucleo- F767ZI	Cortex-M7	1	board	ST-Link (interface stlink- dap.cfg)
nucleo-h743zi	harness (upstream)	ST Nucleo- H743ZI	Cortex-M7	1	board	ST-Link (interface stlink- dap.cfg)
raspberrypi-pico	harness (upstream)	Raspberry Pi Pico (RP2040)	Cortex- M0+	1	board	CMSIS DAP (interface cmsis- dap.cfg)

Every platform has a `-cmake-gcc-debug` and a `-cmake-gcc-release` configuration; native also has `-cmake-sys-*`, `-cmake-gccNN-*` and `-cmake-clangNN-*`. Only these configurations carry per-test `test-<app>-<variant>` actions: the `cortexm-*`, `aarch32-*`, `aarch64-*`, `qemu-cortex-*`, `2xcortex-m33`, `pico2-1cpu`, `native-cmake-gcc-*` and `native-cmake-sys-*` ones (the release configurations inherit them from debug).

The QEMU machine for the Pis is `raspi3b -smp 4`; the AArch32 images are started through a small AArch64 boot shim (`shim8.img`, image at `0x10000`). The Cortex-M `-qemu` images of `cortexm-pico2` and `cortexm-pico2-rp2350b-psram` run on `mps2-an500 -cpu cortex-m7`.

The four `qemu-cortex-*` platforms link the local Cortex-M port's generic single-core QEMU cores (`micro-os-plus::cortexm-qemu-m0/-m3/-m4f/-m7`) and the generic device in `tests/device-qemu-cortexm`, and run the SMP kernel single-core (`OS_NCPU=1`, no `OS_USE_SMP_SCHEDULER`). Each registers three suites: `<platform>-rtos-apis-test`, `-mutex-stress-test` and `-cmsis-os-validator-test`. (`2xcortex-m33` and `pico2-1cpu` likewise link the local generic M33 core, `micro-os-plus::cortexm-qemu-m33`.)

3. Probes

Board	Probe	USB id	Transport, speed	Config
Pi Zero 2 W, Pi 3 B (aarch32 and aarch64)	SEGGER J- Link — default	—	JTAG	test/ boards/rpi- zero-2w/ openocd- jlink- rpi3.cfg
Pi Zero 2 W, Pi 3 B, with <code>PROBE=olimex</code>	Olimex ARM-USB- OCD (FTDI)	—	JTAG, 1000 kHz	test/ boards/rpi- zero-2w/ openocd- olimex.cfg
Luckfox Lyra B	WCH-Link in ARM mode (CMSIS- DAP)	1a86:8011	SWD, 4000 kHz	test/ boards/ luckfox- lyra/ openocd.cfg
Pico 2	any CMSIS- DAP (auto- selected, no vid_pid pinned)	—	SWD, 1000 kHz	test/ boards/ pico2/ openocd.cfg
Pi-Zero RP2350B	XV-Link CMSIS- DAP	0416:5951	SWD, 2000 kHz	test/ boards/ pico2- pizero/ openocd.cfg
WeAct RP2350B + PSRAM	CMSIS- DAP	c251:f001	SWD, 1000 kHz	test/ boards/ pico2- rp2350b- psram/ openocd.cfg
Nucleo- F411RE	on-board ST-Link v2.1	—	SWD, <code>connect_assert_srst</code>	test/ boards/

Board	Probe	USB id	Transport, speed	Config
				nucleof411/ openocd.cfg
WeAct F411CE	DAPLink CMSIS- DAP (override: WCH-Link 1a86:8011 through DAP_VID_PID)	0d28:0204	SWD, 1000 kHz (ADAPTER_KHZ)	test/ boards/ weactf411/ openocd.cfg
WeAct F412RE	ST-Link V2 / V2-1 / V3	by class	SWD (ADAPTER_KHZ)	test/ boards/ weactf412/ openocd.cfg

The Pi 3 B hw.sh is a wrapper that execs the Zero 2 W's, so both Pis take the same PROBE switch. The paths above are relative to the port repository (aarch32 or aarch64 for the Pis and the Lyra, cortexm for the rest).

4. The harness suites (portable)

Their sources live in tests/sources/<suite>/. They have no main(): the entry is os_main(), and its return code is the verdict. On the port boards a per-board wrapper wraps os_main (-Wl,--wrap=os_main), prints the RESULT: line and stops the run. It is each suite's harness-suite.cpp on the Pi ports, and one test/<board>/harness-suite.hpp per board, included by the suites' harness-suite.cpp, on cortexm.

Suite	What it does
rtos-apis	Exercises the µOS++ C++ API, C API, ISO (std::) API, CMSIS-RTOS v1 wrapper and the POSIX I/O layer, plus a Chan FatFs test.
mutex-stress	Several threads take one mutex at random intervals; the test checks that every thread got it and the distribution is sane.
fp-switch	Six unpinned threads at three priorities each load their own pattern into s0-s31 and the FPSCR flags, spin long enough to be preempted (and, on SMP, moved to the other core), and

Suite	What it does
	check the registers read back unchanged. Needs an FPU; without one it passes with nothing to test.
cmsis-os-validator	The Arm CMSIS-RTOS v1 validator (60 test cases: threads, timers, signals, semaphores, mutexes, memory pools, message and mail queues, with interrupt-context cases). On a Cortex-M the interrupt is NVIC IRQ 0; see below for the other cores.
blinky	Blinks the LED. Enabled only on nucleo-f411re (platforms/nucleo-f411re/cmake/definitions.cmake), where it is the case blinky-test.
instrumentation	SEGGER SystemView use cases. Enabled only on nucleo-f411re, which builds instrumentation-test and rtos-apis-instrumentation-test but registers no CTest case for them: they are run from SEGGER Ozone with a J-Link (see tests/sources/instrumentation/README.md).

Where each suite runs:

Platform	rtos-apis	mutex-stress	cmsis-os-validator	Mode
aarch32-rpi-zero-2w, aarch32-rpi3b, aarch64-rpi-zero-2w, aarch64-rpi3b	qemu, hwd	qemu, hwd	qemu, hwd	SMP 4
aarch32-luckfox-lyra	—	hwd (mutex-stress-test)	—	SMP 3
cortexm-pico2	qemu, hwd, ram	qemu, hwd, ram	qemu, hwd, ram	SMP 2 on hwd; single on qemu &

Platform	rtos-apis	mutex-stress	cmsis-os-validator	Mode
				ram (SRAM load)
cortexm-nucleof411, cortexm-weactf411, cortexm-weactf412	hwd	hwd	hwd	SMP ×1
native	host (rtos-apis, single; smp-rtos-apis, SMP 4)	host (mutex-stress, single; smp-mutex-stress, SMP 4)	host	see §8
2xcortex-m33	QEMU	QEMU	QEMU	SMP 2
pico2-1cpu	QEMU	QEMU	QEMU	single
qemu-cortex-m0/m3/m4f/m7f	QEMU	QEMU	QEMU	single (local cortexm port's generic QEMU cores, OS_NCPU=1)
nucleo-*, raspberrypi-pico	yes	yes	yes	single (upstream); nucleo-f411re adds blinky-test

fp-switch runs where an M33 FPU does: cortexm-pico2 (qemu, hwd), 2xcortex-m33 (QEMU, SMP 2) and pico2-1cpu (QEMU, single).

On cortexm-pico2 the -qemu image of a suite does not run the board at all: QEMU has no RP2350, so the image links the generic single-core Cortex-M QEMU core in its place, and the port's builder forces NCPU to 1 for it (<cortexm>/test/CMakeLists.txt).

The validator is CMSIS-RTOS v1, which assumes one CPU. On the SMP platforms the kernel's wrapper pins every thread `osThreadCreate()` makes to core 0 (`src/rtos/os-c-wrapper.cpp`). On the Pi ports the board's `harness-suite.cpp` pins the main thread too; the pico2 wrapper (`test/pico2/harness-suite.hpp`) does not, so there the validator's main thread keeps the default affinity while every thread it creates is on core 0. On the BCM2837 (both Pi ports) its NVIC IRQ 0 is emulated with the local Mailbox 1

interrupt, and the Cortex-M DWT cycle counter it probes is answered by a RAM page that says “no cycle counter”. Both exist only in the validator’s image (U05_CMSIS_05_VALIDATOR). On native, the validator xpack’s own SIGUSR1/SIGALRM shims stand in for the NVIC.

5. AArch32 and AArch64 — Raspberry Pi Zero 2 W and Pi 3 B

The same fifteen test directories exist on all four (port, board) pairs: <port>/test/rpi-zero-2w/ and <port>/test/rpi3b/. The Pi 3 B builds use the Zero 2 W’s board sources with BOARD_RPI3B. Every test is built twice, -qemu and -hwd, except as noted.

Test	What it does	Mode
smp_test0	Phase-1 bring-up: PL011 UART, MMU (cacheable, shareable DRAM), the 1 ms generic- timer tick and the scheduler, with one thread printing 10 heartbeats.	SMP 4, 1 active
smp_test1	Semaphore ping- pong across cores: pinger on core 0, ponger on core 1, a logger on core 2 blinking the LED and printing the round count and cores.	SMP 4
smp_test2	Lock coherency: one worker pinned per core bumps a shared counter under a mutex; the final count must be $OS_NCPU \times ITER$.	SMP 4
smp_test3	message_queue producer (core 0) and consumer (core 1) that tests primes and toggles the LED; a logger prints throughput.	SMP 4
smp_test4		SMP 4

Test	What it does	Mode
	Load balancing: CPU-bound workers with no affinity; a reporter prints each worker's per-core histogram.	
smp-mat-test	Parallel block Gaussian elimination ($N = 120$, $B = 20$) across all cores with spin barriers, a thread- pool demo, and a classical LU solve as ground truth; compares and times both.	SMP 4
smp-mat-sdcard-test	The same solver at $N = 200$ with the matrices stored on and loaded from the SD card (flatfs on disk.img under QEMU; FatFs under /tests on the boot card on hardware).	SMP 4
smp-num-test	Five prioritised threads: UART ticker, LED blinker, an SD text writer, an FP compute writer and a reader, all sharing num.txt under one mutex, paced by a counting semaphore.	SMP 4
smp-pipeline-test	13 threads (4 producers, 4 compute workers, 2 SD writers, an auditor that checks CRC and sequence, an LED pacer, telemetry) with heavy yield().	SMP 4
smp-pro-cons-test	Producer/consumer that uses every kernel object (thread, memory	SMP 4

Test	What it does	Mode
	pool, message queue, both semaphores, mutex, condition variable, event flags, timer, sysclock, suspend/resume).	
sd_test	SD card: under QEMU, sd::SdCard + flatfs on a host-seeded disk.img (host ↔ device round trip); on hardware, FatFs read/write under / tests on the boot card (never formatted).	SMP 4
usb_test	DWC2 USB gadget file transfer: a host tool sends files over bulk OUT, stored to the SD card and hex-dumped to the UART and semihosting, with unpinned workers keeping every core busy.	SMP 4
rtos-apis, mutex-stress, cmsis-os-validator	The harness suites (§4).	SMP 4

Exception: `usb_test` has **no -hwd case on the Pi 3 B**, only `-qemu`. The Pi 3 B's USB ports sit behind the LAN9514 hub, so the DWC2 can only be a host.

Case counts: Zero 2 W 30 (15 × 2), Pi 3 B 29.

6. AArch32 — Luckfox Lyra B (RK3506)

Hardware only: the board declares no QEMU linker script. 19 port tests plus one harness suite, all `-hwd`, all SMP 3.

Test	What it does
smp_test0 ... smp_test4	The same bring-up, ping-pong, lock-coherency, message-

Test	What it does
	queue and load-balancing tests as on the Pis (§5), on three cores.
smp-mat-test, smp-mat-sdcard-test, smp-num-test, smp-pipeline-test, smp-pro-cons-test, sd_test	As on the Pis (§5), on the Lyra's SD card controller (DW-MMC).
smp_test5	Three pinned threads: an SD card thread on core 0 (FatFs mount, formatting FAT32 if none), an LED blinker on core 1 and a logger on core 2.
smp_test6	USB-to-UART logger on three cores: a DWC2 gadget task, a heartbeat logger and an LED task; driven from the host with usb_test_host.
smp_test7	USB vendor gadget receives a file, the SD task writes it as received.dat, then asks on the UART whether to delete it; the LED reflects the answer.
smp_test_int	First external interrupt: UART1 RX on the GIC (ID 99), with an ISR echo and a TX line lock shared with the ISR.
smp_test_int2	Interrupt-driven TX through a THRE ring, and the UART1 interrupt routed to core 1 via GICD_ITARGETSR.
smp_test_int3	PC keyboard → board: USB OTG1 as an interrupt-driven vendor gadget on core 0 (GIC ID discovered at runtime), UART1 interrupt-driven on core 1. Needs the host tool kbd_forward.
smp_test_int4	Keystroke logger with three IRQs on three cores: USB (core 0) → ring → keys.log on the card, written with the DW-MMC data-transfer-over IRQ (core 2); UART on core 1.
smp_test_int5	AMP doorbell: the RK3506's Cortex-M0 fires Mailbox interrupts every ~40 ms to A7

Test	What it does
mutex-stress-test	cores 1 and 2, which report to core 0 over a shared ring. The harness suite, built by the platform itself (not a port test directory).

Every Lyra test ends with a `RESULT:` line and a semihosting exit. The four USB tests (`smp_test6`, `smp_test7`, `smp_test_int3`, `smp_test_int4`) wait up to 120 s for the host to configure the gadget, then fail; they are built `UOS_CONSOLE_UART_ONLY`, so their console does not mirror every line through semihosting (each such trap halts the core, and during USB traffic that lost the WCH-Link’s debug link). Their host tools are C/C++ sources in the test’s own host/ directory: `usb_test_host` for `smp_test6`/`smp_test7`, `kbd_forward` for `smp_test_int3`/`smp_test_int4`.

7. Cortex-M

7.1 RP2350 boards

Three boards on the same silicon (2× Cortex-M33): `pico2`, `pico2-pizero` and `pico2-rp2350b-psram`. On the single-core tests `OS_NCPU` is 1 (`board_test_ncpu`); the “bare” tests link no kernel (`BOARD_TEST_NO_KERNEL`).

Test	What it does	Mode	pico2	pizero	rp2350b-psram
<code>exc-test</code>	Bare exception-entry probe: SysTick with MSP, then PendSV with PSP.	bare	hwd	hwd	hwd
<code>smp-test0</code>	Bare boot chain and dual-core launch: C++ constructors, 150 MHz clocks, UART; core 1 blinks the LED.	bare, 2 cores	hwd	hwd	hwd
<code>smp-test1</code>	Single-core RTOS bring-up: three threads, sysclock, mutex, counting semaphore, LED heartbeat.	single	qemu, hwd	hwd	qemu, hwd
<code>sc-test-ko</code>		single		hwd	

Test	What it does	Mode	pico2	pizero	rp2350b-psram
	Single-core kernel-object test: ten self-checking sub-tests (mutex, semaphores, queues, event flags, timers, ...), controller and helper on core 0.		qemu, hwd		qemu, hwd
smp-test-ko	The SMP twin of sc-test-ko: controller pinned to core 0, helper to core 1, so every hand-off is cross-core.	SMP 2	hwd	hwd	hwd
smp-test2	Dual-core SMP: per-core PendSV and SysTick, affinity (one worker per core), SIO spinlock-backed kernel lock.	SMP 2	hwd	hwd	hwd
smp-test3	Cross-core IPI through the SIO FIFO: semaphore ping-pong with the one-way wake latency measured on TIMER0.	SMP 2	hwd	hwd	hwd
smp-test4	Kernel objects under dual-core contention: seven round-trip sub-tests between a controller (core 0) and a worker (core 1).	SMP 2	hwd	hwd	hwd
smp-test5	First peripheral interrupt: UART0 TX IRQ serviced on core 1, draining a ring filled by core 0.	SMP 2	hwd	hwd	hwd
smp-mat-test	Parallel block solver ($N = 100$, B	SMP 2	hwd	hwd	hwd

Test	What it does	Mode	pico2	pizero	rp2350b-psram
	= 50) on two cores. On pico2-rp2350b-psram its data lives in PSRAM.				
smp-test-usb-cdc-acm	USB CDC-ACM virtual serial port: echoes what the host types, mirrored on the UART.	SMP 2	hwd	hwd	—
smp-test-usb-hid	USB HID keyboard forwarding; keystrokes printed on the UART.	SMP 2	hwd	hwd	—
psram-exec	Bare SRAM supervisor that copies two apps into PSRAM and switches between them on a key press.	bare	—	hwd	—
psram-mat-test-250	smp-mat-test with PSRAM brought up by the app itself, at 250 MHz, from an SRAM-boot linker script.	SMP 2	—	hwd	—
smp-test-nested	Nested interrupts on both cores (SysTick, UART0, TIMER0 IRQ 2/3 at different priorities) with thread migration and a lock-free trace. Passes once both workers have run 20 beats and all three nesting interrupts have fired.	SMP 2	—	—	hwd
smp-test-nested-clock	smp-test-nested plus a clock tree driven from a potentiometer (ADC) and the on-	SMP 2	—	—	hwd

Test	What it does	Mode	pico2	pizero	rp2350b-psram
	die temperature sensor; data in PSRAM, 250 MHz enabled. Interactive: waits for y on the UART (a pot position other than the current clock reboots into it first), then the same 20-beat verdict.				
smp-test-nested-clock_200	The same, without PIC02_ENABLE_250MHZ (the clock tree at its default).	SMP 2	—	—	hwd
smp-test-nested-clock_250	The same, with 250 MHz enabled.	SMP 2	—	—	hwd
rtos-apis, mutex-stress, cmsis-os-validator, fp-switch	The harness suites (§4).	SMP 2 (hwd); single (qemu)	qemu, hwd	—	—
cmsis-os-validator-ram, mutex-stress-ram, rtos-apis-ram	RAM-resident harness suites executing entirely from internal SRAM (no flash writes).	single	hwd	—	—

Case counts: pico2 25, pico2-pizero 14, pico2-rp2350b-psram 16.

7.2 STM32F4 boards (one CPU)

cortexm-nucleof411, cortexm-weactf411 and cortexm-weactf412, all hardware only. The harness suites build the SMP scheduler at OS_NCPU = 1; the board's own test uses the kernel's non-SMP branch.

Test	What it does	Mode	nucleof411	weactf411	weactf412
mos-test1	Single-CPU acceptance:	single	hwd	hwd	hwd

Test	What it does	Mode	nucleof411	weactf411	weactf412
	producers and consumers over a bounded buffer with plain and recursive mutexes, counting semaphores, sysclock and an LED thread.				
spi-pipeline	SPI filesystem pipeline (ported from a Rust example): five threads, eight semaphores, Base64 encode to partition 1, decode and CRC check to partition 2.	single	—	hwd	—
uart-test1	USART1 echo: RX interrupt → bounded buffer → TX via DMA2, with an LED thread.	single	—	—	hwd
rtos-apis, mutex-stress, cmsis-os-validator	The harness suites (§4).	SMP ×1	hwd	hwd	hwd

8. POSIX — native host

micro-os-plus-iii-posix-arch.git/test/native/. Host threads stand in for CPUs: NCPU defaults to 4 (-NCPU=1 builds everything single-core). All cases are -host.

Test	What it does	Mode
smp_test0 ... smp_test4	The Pi tests of the same names (§5), on host threads.	SMP 4 (smp_test0: 1 active)
smp-mat-test, smp-num-test, smp-pipeline-test, smp-pro-cons-test	As on the Pis; the SD card is flatfs over a host file.	SMP 4
mutex-stress	The upstream suite, unchanged.	single
rtos-apis	The upstream suite, with the POSIX I/O layer.	single
smp-mutex-stress	The SMP leg of mutex-stress, with smp_test2's start-up.	SMP 4
smp-rtos-apis	The whole rtos-apis sequence once per core, from a driver pinned to that core, with the API threads unpinned; fails unless every core did the work.	SMP 4
flatfs-test	flatfs on a host-file card: append 64 KiB to a file created empty, then check it, a file beside it and a fresh mount.	SMP 4
mutex-ceiling-test	A thread above a robust protect mutex's ceiling gets EINVAL; after it is destroyed, lock/unlock at the ceiling must still work (no EOWNERDEAD).	SMP 4
cmsis-os-validator	The validator (§4), registered by the	SMP 4, validator on core 0

Test	What it does	Mode
	harness platform itself.	

16 cases.

9. How to run

From `micro-os-plus-iii-smp.git/tests`, for a configuration `C` such as `aarch64-rpi3b-cmake-gcc-debug`:

```
xpm run install --config C && xpm run prepare --config C && xpm
    run build --config C
xpm run test --config C                                # every
    qemu/host case, no hwd
xpm run test-<app>-<variant> --config C                 # one case,
    builds first
PATH="$PWD/build/C/xpacks/.bin:$PATH" ctest --test-dir build/C
    -N # list the cases
```

For a `-hwd` case, power-cycle the board first and run exactly one case. On the Pis choose the probe with `PROBE=jlink|olimex`:

```
cd build/C && PROBE=olimex PATH="$PWD/xpacks/.bin:$PATH" ctest -V
    -R <platform>-<app>-hwd
```